

Rectified Schrödinger Bridge Matching for Few-Step Visual Navigation

Wuyang Luan, Junhui Li, Weiguang Zhao, Wenjian Zhang, Tieru Wu[†], and Rui Ma[†]

Abstract—Visual navigation from monocular image streams requires an agent to transform high-dimensional visual observations into continuous, long-horizon action trajectories. Generative policies based on diffusion models and Schrödinger Bridges (SB) can model multimodal action distributions faithfully, yet they require dozens of integration steps due to high-variance stochastic transport. We propose Rectified Schrödinger Bridge Matching (RSBM), a visual-conditioned trajectory generation framework that exploits a shared velocity-field structure between standard Schrödinger Bridges ($\varepsilon = 1$) and deterministic Optimal Transport ($\varepsilon \rightarrow 0$), governed by a single entropic regularization parameter ε . We prove two theoretical results: (1) the conditional velocity field maintains the same functional form across the entire ε spectrum, so one network parameterization serves all regularization strengths; and (2) reducing ε linearly decreases the conditional velocity variance, which improves the stability of coarse-step ODE integration. Anchored to a learned conditional prior that shortens transport distance, RSBM achieves over 94% cosine similarity and 92% success rate in merely 3 integration steps, without distillation or multi-stage training, substantially narrowing the gap between high-fidelity generative policies and real-time multimedia systems.

Index Terms—Visual navigation, monocular image streams, visual trajectory generation, Schrödinger bridge, flow matching, diffusion models, few-step inference, optimal transport.

I. INTRODUCTION

VISUAL navigation is a fundamental perception-and-decision problem in intelligent multimedia systems. An agent must interpret streaming visual observations, preserve scene semantics over time, and convert them into reliable action trajectories. Recent generative policies powered by Denoising Diffusion Probabilistic Models (DDPMs) have demonstrated clear advantages over deterministic regression in this setting, because they can faithfully represent the multimodal nature of future actions under visual ambiguity. When multiple valid continuations exist for the same observation, deterministic regressors average across modes and produce infeasible trajectories, whereas diffusion-based policies maintain distinct modes and generate physically plausible plans.

Despite these representational advantages, deploying diffusion-based policies in real-time multimedia systems

presents a critical practical challenge in inference latency. Standard diffusion models and Schrödinger Bridges (SB) construct high-variance Brownian transport paths from noise to data, requiring dozens of iterative denoising steps for accurate trajectory recovery. For vision-driven navigation where trajectory prediction must operate on top of continuous visual perception under tight response budgets, such computational overhead is prohibitive. Existing acceleration strategies, including step-skipping schedules, knowledge distillation, and progressive training, can reduce the number of function evaluations but typically sacrifice trajectory fidelity or require costly multi-stage procedures.

A separate line of work addresses the initialization problem. Rather than starting from uninformative isotropic Gaussian noise, prior-conditioned methods such as NaviBridger anchor the generative process to a learned motion prior, which shortens the effective transport distance and improves convergence. However, even with a good prior, the underlying Brownian Bridge still exhibits high path curvature and variance in the interior of the transport interval, limiting the achievable quality at very low step counts.

To address both challenges simultaneously, we propose Rectified Schrödinger Bridge Matching (RSBM). Our key insight is that standard SB with $\varepsilon = 1$ and the linear interpolants underlying Conditional Flow Matching with $\varepsilon \rightarrow 0$ are *endpoints of the same entropic regularization spectrum*. By introducing $\varepsilon \in (0, 1]$ into the bridge kernel, RSBM provides a continuous interpolation between maximum-entropy stochastic transport and deterministic optimal transport. We prove that the conditional velocity field maintains the same functional form across the entire ε family (Theorem 1), and that reducing ε linearly reduces velocity variance (Proposition 1). Combined with a learned conditional prior that shortens transport distance, these properties enable RSBM to generate high-fidelity trajectories in as few as 3 ODE steps while preserving multimodal behavior.

In summary, our main contributions are:

- **A Continuous SB–FM Interpolation:** We establish that ε -parameterized bridge kernels form a continuum from maximum-entropy Schrödinger Bridges to deterministic optimal transport, and prove that the conditional velocity field is structurally invariant across this spectrum (Theorem 1), enabling a shared network parameterization.
- **Variance Reduction with Theoretical Guarantee:** We prove that ε -rectification linearly reduces conditional velocity variance (Proposition 1), with full derivations from Schrödinger Bridge theory in Section IV.

[†]Corresponding authors (e-mail: wutr@jlu.edu.cn; ruim@jlu.edu.cn).

W. Luan, T. Wu, and R. Ma are with the School of Artificial Intelligence, Jilin University, Changchun 130012, China (e-mail: luanwy25@mails.jlu.edu.cn).

J. Li is with the College of Computer Science, Chongqing University, Chongqing 400044, China (e-mail: junhuili@stu.cqu.edu.cn).

W. Zhao is with the Department of Computer Science, University of Liverpool, Liverpool L69 3BX, UK (e-mail: weiguang.zhao@liverpool.ac.uk).

W. Zhang is with Changchun GenY Technology Co., Ltd., Changchun, China (e-mail: zhangwenjian@genycc.cn).

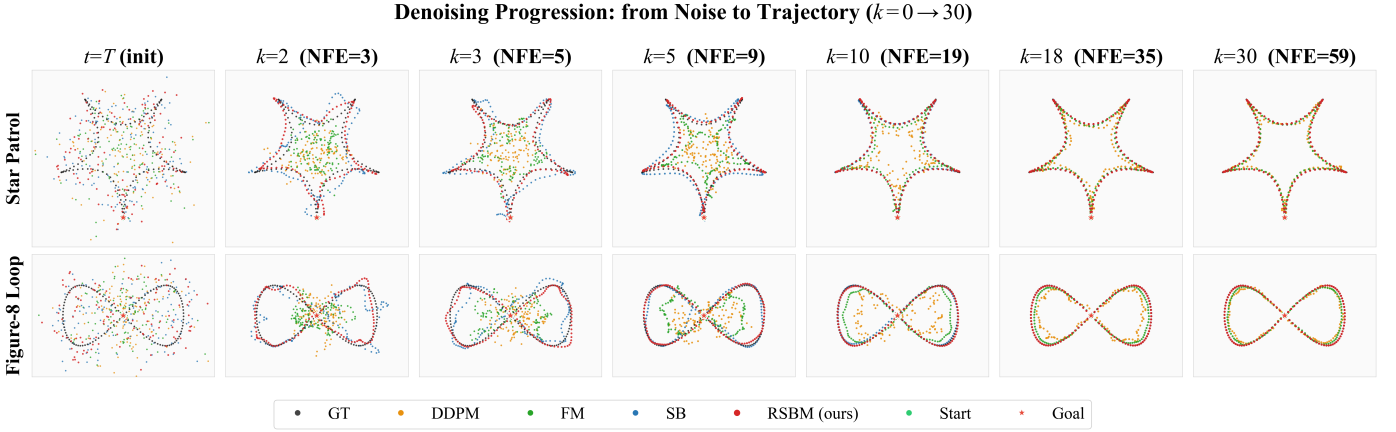


Fig. 1. Denoising progression on two toy trajectories (star patrol and figure-8 loop). At $t=T$ all methods start as unstructured point clouds (Gaussian noise for DDPM/FM; learned prior for NaviBridger/RSBM). By $k=2$ (NFE=3), only RSBM closely matches the GT, while baselines require $k \geq 10$ to converge. At $k=30$ (NFE=59), all methods have converged, confirming the quantitative trends in Table III.

- **Single-Stage Few-Step Performance:** RSBM achieves $6.3 \times$ lower MSE than NaviBridger at $k=3$ and matches its $k=10$ accuracy with $3.8 \times$ fewer function evaluations, without distillation or multi-stage training as in [1], [2].

II. RELATED WORK

Generative Policies for Visual Navigation. Visual navigation has evolved from classical modular pipelines [3], [4] and end-to-end approaches based on reinforcement learning [5], [6] or behavioral cloning [7], [8] toward generative policies. Foundation models such as ViNT [9] demonstrate strong sample efficiency, while 3D representations like Gaussian Splatting [10], [11] improve spatial grounding. However, deterministic planners struggle in multimodal environments where averaging across distinct valid modes yields infeasible plans [12], [13]. Score-based diffusion and flow matching models [14]–[18] address this by capturing multimodal action distributions; NoMaD [19] was among the first to apply diffusion to navigation. A key limitation remains that initialization from isotropic Gaussian noise [20], [21] necessitates long reverse processes, limiting real-time deployment.

Diffusion Bridges and Flow Matching. Diffusion bridges [22] generalize endpoint-conditioned processes [23] via Doob’s h -transform [24], with extensions to discrete-time [25] and continuous-time formulations [26]. Recent Schrödinger Bridge Matching methods [27], [28] learn bridges without iterative simulation; Light and Optimal SBM [29] and Adversarial SBM [30] further improve training efficiency. In navigation, NaviBridger [31] initializes a diffusion bridge from a learned motion prior, NaviD [32] leverages depth constraints, FlowNav [33] combines CFM with depth priors, and StepNav [34] constructs geometry-aware priors online. However, standard bridge formulations exhibit ill-conditioned dynamics near temporal boundaries, introducing truncation errors in few-step regimes [35], [36]. Our method addresses this by explicitly constraining bridge variance via a single ε parameter, yielding more stable transport dynamics.

Path Straightening and Accelerated Sampling. Rectified Flow [2] straightens ODE trajectories via iterative reflow;

Consistency Models [1] distill pre-trained diffusion models into few-step generators. Both require multi-stage training. RSBM achieves path straightening in a single stage by explicit variance control from an informed prior, retaining multimodal coverage at intermediate ε .

III. METHOD

Our framework, illustrated in Fig. 2, consists of three tightly coupled components. A dual-stream vision encoder f_ϕ compresses streaming observations and the goal image into a context vector $\mathbf{c}$ (§III-A). A learned variational prior network g_ψ then maps $\mathbf{c}$ to a coarse action initialization $\mathbf{a}_T$ (§III-A). Finally, a conditional velocity network $\mathbf{v}_\theta$ drives an ε -rectified Schrödinger Bridge that progressively refines $\mathbf{a}_T$ into the output trajectory $\mathbf{a}_0$ in as few as 3 ODE steps (§III-B–III-D).

A. Problem Formulation for Visual Navigation

We formulate visual navigation as a conditional generative modeling problem. At each decision step, the agent receives a streaming observation sequence $\mathcal{O} = \{I_{t-C}, \dots, I_t\}$ of monocular RGB images and a goal image I_g , and must produce an action trajectory $\mathbf{a}_0 \in \mathbb{R}^{H \times 2}$ representing H future waypoints in local coordinates. A perception encoder maps visual inputs into a compact context vector:

$$\mathbf{c} = f_\phi(\mathcal{O}, I_g) \in \mathbb{R}^d \quad (1)$$

Rather than generating $\mathbf{a}_0$ from uninformative Gaussian noise, a scheme that typically requires many denoising steps, we anchor the generative process to an informative conditional prior. A learned variational prior network produces a structured initialization $\mathbf{a}_T$ that coarsely captures navigational intent:

$$\mathbf{a}_T = g_\psi(\mathbf{z}, \mathbf{c}), \quad \mathbf{z} \sim q_\psi(\mathbf{z} \mid \mathbf{c}, \mathbf{a}_0) \text{ (train)} / \mathcal{N}(\mathbf{0}, \mathbf{I}) \text{ (test)} \quad (2)$$

This prior $\mathbf{a}_T$ serves as the terminal boundary condition ($t=T$) for our diffusion bridge: the model need only learn the *residual refinement* from a coarse prior to the precise ground-truth trajectory, substantially shortening the effective transport distance.

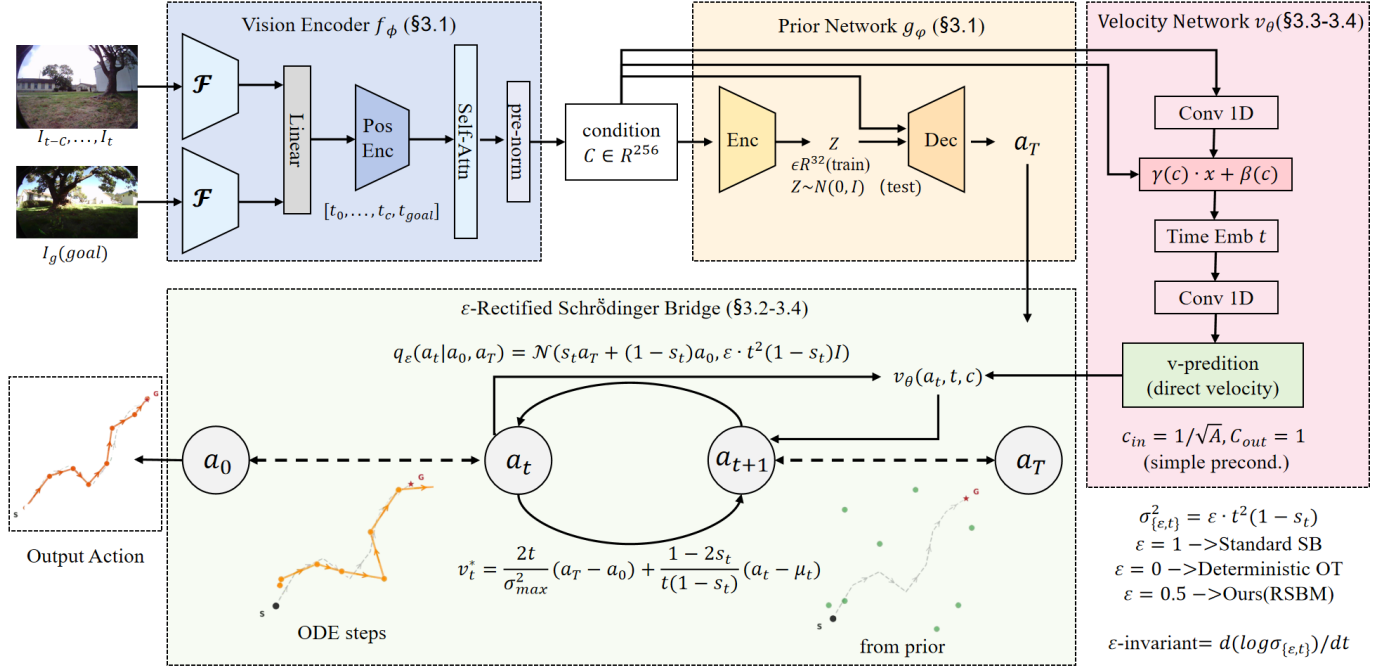


Fig. 2. **Overview of the RSBM framework.** *Left:* A dual-stream EfficientNet-B0 vision encoder f_ϕ (§III-A) extracts observation and goal features, which are fused via positional encoding and self-attention into a context vector $\mathbf{c} \in \mathbb{R}^{256}$. *Center:* A learned variational prior network g_ϕ (§III-A) produces a coarse action prior $\mathbf{a}_T$. *Right:* A conditional U-Net 1D velocity network $\mathbf{v}_\theta$ (§III-C) with FiLM conditioning iteratively refines $\mathbf{a}_T$ into the output trajectory $\mathbf{a}_0$ via an ε -Rectified Schrödinger Bridge (§III-B–III-D). The three trajectory snapshots illustrate the progressive refinement: from scattered prior waypoints (green), through partially converged intermediate states (orange), to the final high-fidelity trajectory (red) closely matching the ground truth (dashed gray).

B. The ε -Rectified Conditional Bridge Kernel

Standard Diffusion Bridge Models construct a Brownian Bridge between $\mathbf{a}_0$ and $\mathbf{a}_T$ with variance $\sigma_t^2 = t^2(1 - t^2/\sigma_{\max}^2)$. These high-variance stochastic paths entangle intermediate states and necessitate many integration steps for accurate recovery. To rectify this, we introduce a structural regularization parameter $\varepsilon \in (0, 1]$ that explicitly controls the path geometry. The forward transition kernel of our rectified bridge is:

$$q_\varepsilon(\mathbf{a}_t | \mathbf{a}_0, \mathbf{a}_T) = \mathcal{N}(\boldsymbol{\mu}_t, \sigma_{\varepsilon,t}^2 \mathbf{I}) \quad (3)$$

where the mean seamlessly interpolates between data and prior:

$$\boldsymbol{\mu}_t = s_t \mathbf{a}_T + (1 - s_t) \mathbf{a}_0, \quad s_t = \frac{t^2}{\sigma_{\max}^2} \quad (4)$$

and the variance is scaled by ε to enforce trajectory straightness:

$$\sigma_{\varepsilon,t}^2 = \varepsilon \cdot t^2(1 - s_t) \quad (5)$$

Boundary conditions. For any $\varepsilon \in (0, 1]$, we have $\sigma_{\varepsilon,0}^2 = 0$ (exact data recovery) and $\sigma_{\varepsilon,\sigma_{\max}}^2 = 0$ (exact prior matching), ensuring the bridge correctly pins both endpoints.

Geometric interpretation. Setting $\varepsilon = 1$ recovers the standard Brownian Bridge. As $\varepsilon \rightarrow 0$, the kernel collapses to $\delta(\mathbf{a}_t - \boldsymbol{\mu}_t)$, the deterministic displacement interpolant of Monge–Kantorovich optimal transport. Intermediate values smoothly trade off between stochastic diversity and path straightness. This formulation is grounded in entropic optimal transport; the complete derivation from stochastic optimal control principles is in Section IV.

C. Conditional Velocity Field and Flow Matching Objective

To enable scalable training, we derive the conditional velocity field of the rectified bridge and formulate a simulation-free Flow Matching objective. Using the reparameterization $\mathbf{a}_t = \boldsymbol{\mu}_t + \sigma_{\varepsilon,t} \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the target velocity is:

$$\mathbf{v}_t^*(\mathbf{a}_t | \mathbf{a}_0, \mathbf{a}_T) = \frac{d\boldsymbol{\mu}_t}{dt} + \frac{d \log \sigma_{\varepsilon,t}}{dt} (\mathbf{a}_t - \boldsymbol{\mu}_t) \quad (6)$$

Computing the key derivatives (complete algebra in Section IV-F):

$$\frac{d\boldsymbol{\mu}_t}{dt} = \frac{2t}{\sigma_{\max}^2} (\mathbf{a}_T - \mathbf{a}_0) \quad (7)$$

$$\frac{d \log \sigma_{\varepsilon,t}}{dt} = \frac{1 - 2s_t}{t(1 - s_t)} \quad (8)$$

Theorem 1 (Velocity Structure Invariance). For the ε -rectified bridge family $\{q_\varepsilon\}_{\varepsilon \in (0,1]}$ (Eq. 3), the logarithmic derivative of the standard deviation satisfies $d \log \sigma_{\varepsilon,t}/dt = (1 - 2s_t)/[t(1 - s_t)]$, which is *independent of ε* . Consequently, the functional form of the conditional velocity $\mathbf{v}_t^*$ (Eq. 6) is invariant across the entire ε -spectrum: a single velocity network parameterization is valid for all $\varepsilon \in (0, 1]$.

Proof. Since $\sigma_{\varepsilon,t} = \sqrt{\varepsilon} t \sqrt{1 - s_t}$, the derivative is $d\sigma_{\varepsilon,t}/dt = \sqrt{\varepsilon}(1 - 2s_t)/\sqrt{1 - s_t}$. Their ratio is $\frac{d\sigma_{\varepsilon,t}/dt}{\sigma_{\varepsilon,t}} = \frac{1 - 2s_t}{t(1 - s_t)}$, where $\sqrt{\varepsilon}$ cancels exactly. Full derivation in Section IV-F. $\square$

Theorem 1 shows that SB ($\varepsilon = 1$) and FM ($\varepsilon \rightarrow 0$) share the same conditional velocity functional form: ε controls the *support* of training samples $\mathbf{a}_t$ via $\sigma_{\varepsilon,t}$ (Eq. 5), while leaving the velocity field’s structure intact. In practice, ε acts as a

spatial support constrictor, yielding a lower-variance learning problem when $\varepsilon < 1$ (Proposition 1).

We parameterize a neural velocity network $\mathbf{v}_\theta$, implemented as a Conditional U-Net 1D with FiLM conditioning on $\mathbf{c}$, and train via the simulation-free Conditional Flow Matching loss:

$$\mathcal{L}_{\text{RSBM}} = \mathbb{E}_{t, \mathbf{a}_0, \mathbf{a}_T, \epsilon} \left[\|\mathbf{v}_\theta(\mathbf{a}_t, t, \mathbf{c}) - \mathbf{v}_t^*\|^2 \right] \quad (9)$$

Crucially, v -prediction directly parameterizes the ODE velocity field, avoiding the score-to-drift conversion required by x_0 -prediction approaches. We empirically show this yields 35.6% lower error at $k = 3$; see Section V-E.

D. Few-Step ODE Inference and Error Analysis

At inference, action trajectories are generated by solving the Probability Flow ODE from $\mathbf{a}_T$ to $\mathbf{a}_0$:

$$d\mathbf{a}_t = \mathbf{v}_\theta(\mathbf{a}_t, t, \mathbf{c}) dt \quad (10)$$

using a second-order Heun solver over a Karras timestep schedule $\{t_0 > t_1 > \dots > t_k = 0\}$:

$$\begin{aligned} \mathbf{d}_1 &= \mathbf{v}_\theta(\mathbf{a}_{t_i}, t_i, \mathbf{c}), & \tilde{\mathbf{a}}_{t_{i+1}} &= \mathbf{a}_{t_i} + \mathbf{d}_1(t_{i+1} - t_i) \\ \mathbf{d}_2 &= \mathbf{v}_\theta(\tilde{\mathbf{a}}_{t_{i+1}}, t_{i+1}, \mathbf{c}), & \mathbf{a}_{t_{i+1}} &= \mathbf{a}_{t_i} + \frac{1}{2}(\mathbf{d}_1 + \mathbf{d}_2)(t_{i+1} - t_i) \end{aligned} \quad (11)$$

$$(12)$$

Each step requires two function evaluations; the final evaluation of one step is reused as the first evaluation of the next (FSAL), yielding NFE = $2k - 1$ for k steps.

Why does $\varepsilon < 1$ enable fewer steps? We formalize this via the following result.

Proposition 1 (Velocity Variance Reduction). For the ε -rectified bridge kernel q_ε (Eq. 3), the conditional variance of the target velocity field satisfies:

$$\text{Var}[\mathbf{v}_t^* | \mathbf{a}_0, \mathbf{a}_T] = \varepsilon \cdot \frac{(1 - 2s_t)^2}{1 - s_t} \cdot \mathbf{I}_D \quad (13)$$

where $D = 2H$ denotes the flattened trajectory dimension. In particular, reducing ε linearly reduces the stochastic variation of the velocity field over the training distribution.

Proof. From Eq. (6), the stochastic component of $\mathbf{v}_t^*$ is $\frac{d \log \sigma_{\varepsilon, t}}{dt} \cdot \sigma_{\varepsilon, t} \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The per-component variance is $\left(\frac{d \log \sigma_{\varepsilon, t}}{dt} \right)^2 \cdot \sigma_{\varepsilon, t}^2$. Substituting Eqs. (5) and (8): $\frac{(1-2s_t)^2}{t^2(1-s_t)^2} \cdot \varepsilon t^2(1-s_t) = \varepsilon \cdot \frac{(1-2s_t)^2}{1-s_t}$. $\square$

Proposition 1 has two practical implications. *First*, lower velocity variance means the training target varies less across the bridge support, enabling the network to achieve better approximation quality with the same capacity. *Second*, in practice, smaller ε keeps trajectories closer to the interpolant $\boldsymbol{\mu}_t$, producing lower-curvature paths that are easier to integrate with finite-step solvers. Section IV-H provides the corresponding sampling-error decomposition and its direct connection to ε .

IV. THEORETICAL ANALYSIS

This section derives the theoretical foundations connecting stochastic optimal control, Schrödinger Bridges, and our ε -rectified formulation.

A. Generative Modeling as Stochastic Optimal Control

We cast trajectory generation as a Stochastic Optimal Control (SOC) problem. The goal is to find an optimal control u_t steering a base distribution towards a structured prior while minimizing path cost:

$$\min_u \mathbb{E}_{X \sim p^u} \left[\int_0^1 \frac{1}{2} \|u_t(X_t)\|^2 dt + g(X_1) \right] \quad (14)$$

$$\text{s.t. } dX_t = (f_t(X_t) + \sigma_t u_t(X_t)) dt + \sigma_t dW_t \quad (15)$$

with $X_0 \sim p_{\text{data}}$. By the Hamilton-Jacobi-Bellman equation, the optimal control satisfies $u_t^*(x) = -\sigma_t \nabla V_t(x)$, where V_t is the value function. Under a memoryless relaxation the terminal cost simplifies to $g(x) = \log(p_{\text{base}}^1(x)/p_{\text{prior}}(x))$, connecting SOC to entropic regularized transport.

B. Duality with Schrödinger Bridges

The Schrödinger Bridge framework recasts this SOC problem as an entropy-regularized optimal transport. Under SB optimality, dynamics are governed by coupled Schrödinger potentials ϕ_t and $\hat{\phi}_t$, unified with SOC via the Hopf-Cole transform $\phi_t(x) = \exp(-V_t(x))$. The resulting conditional bridge distribution is Gaussian:

$$q(X_t | X_0, X_1) = \mathcal{N}((1-s_t)X_0 + s_tX_1, t^2(1-s_t)\mathbf{I}) \quad (16)$$

with $s_t = t^2/\sigma_{\text{max}}^2$. The variance $t^2(1-s_t)$ controls stochastic transport diffusivity: it vanishes at both endpoints (exact pinning) and peaks at the midpoint, where transport is most diffuse. This motivates our ε -rectification, which directly scales this variance.

C. Connection to the ε -Rectified Kernel

Building on the bridge kernel derived in Section IV-B, our ε -rectification replaces the variance $t^2(1-s_t)$ with:

$$\sigma_{\varepsilon, t}^2 = \varepsilon \cdot t^2(1-s_t), \quad \varepsilon \in (0, 1] \quad (17)$$

This modulates the entropic regularization strength of the Schrödinger Bridge [37]. Specifically, the SB problem with entropic cost $\gamma \text{KL}(p \| p_{\text{ref}})$ produces a family of solutions indexed by γ ; our ε plays a role analogous to γ/γ_0 where γ_0 is the reference regularization. This interpolates between two extremes:

- $\varepsilon = 1$: Standard Brownian Bridge (maximum entropy, full stochastic transport).
- $\varepsilon \rightarrow 0$: Deterministic displacement interpolant $\boldsymbol{\mu}_t$ (minimum entropy, Monge OT map).

By choosing $\varepsilon \in (0, 1)$, we retain the bridge structure and boundary conditions while concentrating probability mass near the geodesic connecting $\mathbf{a}_0$ and $\mathbf{a}_T$. This provides a principled mechanism for trading off generation diversity against path straightness.

1) *Formal KL Divergence Connection:* We now rigorize the entropic regularization interpretation. Consider the Schrödinger Bridge problem with regularization strength $\gamma > 0$:

$$\min_{p \in \mathcal{P}(\Omega)} \int c(\omega) dp(\omega) + \gamma \text{KL}(p \| p_{\text{ref}}) \quad (18)$$

where $c(\omega) = \frac{1}{2} \int_0^T \|u_t\|^2 dt$ is the kinetic energy cost, p_{ref} is the Brownian Bridge reference, and Ω is the path space. The solution satisfies $p^* \propto p_{\text{ref}} \cdot \exp(-c/\gamma)$, with conditional kernel:

$$p^*(\mathbf{a}_t | \mathbf{a}_0, \mathbf{a}_T) = \mathcal{N}\left(\boldsymbol{\mu}_t, \frac{\gamma}{\gamma_0} t^2 (1-s_t) \mathbf{I}\right) \quad (19)$$

where γ_0 is the reference regularization yielding unit bridge variance. Identifying $\varepsilon = \gamma/\gamma_0$, we obtain $\sigma_{\varepsilon,t}^2 = \varepsilon \cdot t^2 (1-s_t)$, recovering Eq. (17). The KL cost between the ε -rectified and standard bridges admits a closed form:

$$\begin{aligned} \text{KL}(q_\varepsilon(\mathbf{a}_t | \mathbf{a}_0, \mathbf{a}_T) \| q_1(\mathbf{a}_t | \mathbf{a}_0, \mathbf{a}_T)) \\ = \frac{D}{2} (\varepsilon - 1 - \log \varepsilon) \end{aligned} \quad (20)$$

where D is the action space dimensionality. This is strictly non-negative for $\varepsilon \neq 1$ and monotonically increases as $\varepsilon \rightarrow 0$, quantifying the information cost of rectification. For our default $\varepsilon = 0.5$ with $D = 16$ (8 waypoints $\times$ 2 dimensions), $\text{KL} = 8 \times (0.5 - 1 - \log 0.5) = 1.55$ nats, a moderate cost that substantially reduces transport curvature while preserving sufficient stochasticity for multimodal generation.

The boundary conditions $\sigma_{\varepsilon,0}^2 = \sigma_{\varepsilon,\sigma_{\text{max}}}^2 = 0$ hold for any $\varepsilon > 0$, ensuring exact endpoint pinning (see Section III-B).

D. Probability Flow ODE Derivation

The marginal-preserving PF-ODE [38] replaces stochastic dynamics with a deterministic flow. For our ε -rectified bridge, substituting the conditional score $\nabla_{\mathbf{a}_t} \log p(\mathbf{a}_t | \mathbf{a}_0, \mathbf{a}_T) = -(\mathbf{a}_t - \boldsymbol{\mu}_t)/\sigma_{\varepsilon,t}^2$ yields the velocity field in Eq. (6). Since $d \log \sigma_{\varepsilon,t}/dt$ is ε -independent (Section IV-F), the same functional form applies for all ε ; only the training input distribution changes. Inference integrates backward via Heun's method with k steps ($\text{NFE} = 2k-1$).

E. v -Prediction: Signal-to-Noise Ratio Analysis

The velocity field $\mathbf{v}_t^*$ admits three reparameterizations. **ε -prediction** predicts noise $\hat{\varepsilon}$; near $t \approx 0$, small errors are amplified. **x_0 -prediction** predicts $\hat{\mathbf{a}}_0$; near $t \approx \sigma_{\text{max}}$, SNR is low. **v -prediction (Ours)** directly predicts velocity, naturally balancing drift and stochastic terms:

$$\text{SNR}_v(t) = \frac{\|d\boldsymbol{\mu}_t/dt\|^2}{\varepsilon(1-2s_t)^2/(1-s_t)} \quad (21)$$

This ratio is well-behaved across $t \in (0, \sigma_{\text{max}})$, avoiding boundary singularities of both alternatives and directly minimizing ODE integration error.

Algorithm 1: RSBM Training

Input: Dataset $\mathcal{D}$, σ_{max} , ε , learning rate η
repeat
 1. Sample $(\mathcal{O}, I_g, \mathbf{a}_0) \sim \mathcal{D}$; $\mathbf{c} = f_\phi(\mathcal{O}, I_g)$
 2. $\mathbf{a}_T = g_\psi(\mathbf{c}, \mathbf{z})$, $\mathbf{z} \sim q_\psi(\mathbf{z} | \mathbf{c}, \mathbf{a}_0)$; $t \sim \mathcal{U}(\sigma_{\text{min}}, \sigma_{\text{max}})$
 3. $s_t = t^2/\sigma_{\text{max}}^2$; $\boldsymbol{\mu}_t = (1-s_t)\mathbf{a}_0 + s_t\mathbf{a}_T$
 4. $\mathbf{a}_t = \boldsymbol{\mu}_t + \sqrt{\varepsilon} t \sqrt{1-s_t} \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 5. $\mathbf{v}_t^* = \frac{2t}{\sigma_{\text{max}}^2}(\mathbf{a}_T - \mathbf{a}_0) + \frac{1-2s_t}{t(1-s_t)}(\mathbf{a}_t - \boldsymbol{\mu}_t)$
 6. $(\phi, \psi, \theta) \leftarrow (\phi, \psi, \theta) - \eta \nabla[\|\mathbf{v}_\theta(\mathbf{a}_t, t, \mathbf{c}) - \mathbf{v}_t^*\|^2 + \mathcal{L}_{\text{prior}}]$
until converged

Algorithm 2: RSBM Inference

Input: $\mathcal{O}$, I_g , networks $(f_\phi, g_\psi, \mathbf{v}_\theta)$, steps k
 1. $\mathbf{c} = f_\phi(\mathcal{O}, I_g)$; $\mathbf{a}_{t_0} = g_\psi(\mathbf{c}, \mathbf{z})$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 2. **for** $i = 0, \dots, k-1$: Heun step with $\mathbf{v}_\theta$
 $\mathbf{d}_1 = \mathbf{v}_\theta(\mathbf{a}_{t_i}, t_i, \mathbf{c})$; $\tilde{\mathbf{a}} = \mathbf{a}_{t_i} - \Delta t \cdot \mathbf{d}_1$
 $\mathbf{d}_2 = \mathbf{v}_\theta(\tilde{\mathbf{a}}, t_{i+1}, \mathbf{c})$; $\mathbf{a}_{t_{i+1}} = \mathbf{a}_{t_i} - \frac{\Delta t}{2}(\mathbf{d}_1 + \mathbf{d}_2)$
 3. **return** $\hat{\mathbf{a}}_0 = \mathbf{a}_{t_k}$ ($\text{NFE} = 2k-1$)

F. Full Velocity Field Derivation

The four-step derivation (mean derivative, standard-deviation derivative, logarithmic derivative with ε cancellation, and final velocity assembly) is given in the supplementary material (Section II-A). The key result is that $d \log \sigma_{\varepsilon,t}/dt = (1-2s_t)/[t(1-s_t)]$, where $\sqrt{\varepsilon}$ cancels exactly, confirming the ε -invariance stated in Theorem 1.

G. Discussion of Proposition 1

The variance $V(s_t) = \varepsilon \cdot (1-2s_t)^2/(1-s_t)$ vanishes at $s_t = 1/2$ and is bounded by ε near $s_t \rightarrow 0$. Reducing ε to 0.5 halves this variance uniformly, (i) reducing regression difficulty and (ii) making the ODE smoother for finite-step solvers. This explains why the MSE gap between standard SB and RSBM is most pronounced at low k .

H. Sampling Error Analysis

For a k -step Heun solver, the 2-Wasserstein distance satisfies $W_2(\hat{p}_0, p_0) \leq C_1 T \delta + C_2 L_\theta T^3/k^2$ [39], where $\delta^2 = \mathbb{E}\|\mathbf{v}_\theta - \mathbf{v}^*\|^2$. By Proposition 1, lowering ε reduces target variance and hence both δ and the discretization error, consistent with Fig. 3.

I. Training and Inference Algorithms

V. EXPERIMENTS

A. Experimental Setup

Datasets. We evaluate on five public navigation datasets (HuRoN, Recon, SACSOn, SCAND, GoStanford; $\sim 60k$ trajectories total) [9], [19], plus a Gazebo-based *Custom Indoor* (500/100 episodes) and outdoor *CitySim* (400/80). The agent receives 96×96 RGB images and predicts 8-step waypoints. $\varepsilon = 0.5$ is selected on Custom Indoor and fixed for all environments.

Baselines & Metrics. We compare against ViNT (1-shot), NoMaD ($k=20$), DDPM ($k=50$), CFM ($k=10$) [39], and NaviBridger ($k=10$) [31], evaluating MSE $\downarrow$, CosSim $\uparrow$, FDE $\downarrow$,

TABLE I

COMPREHENSIVE COMPARISON. EACH METHOD IS SHOWN AT ITS DEFAULT k AND AT $k=3$, REVEALING DEGRADATION UNDER ZERO-SHOT STEP REDUCTION. GRAY ROWS : $k=3$ VARIANTS. RSBM IS SHOWN AT BOTH $k=3$ AND $k=10$ TO DEMONSTRATE EARLY SATURATION. **BOLD**: BEST OVERALL PER COLUMN.

Method	k	NFE	Custom Indoor					CitySim (Outdoor)				
			MSE↓	CosSim↑	FDE↓	Col.%↓	Suc.%↑	MSE↓	CosSim↑	FDE↓	Col.%↓	Suc.%↑
ViNT [9] (1-shot)	1	1	6.50	0.720	2.85	1.58	28	8.20	0.650	4.50	0.41	38
NoMaD [19]	20	20	3.60	0.820	1.95	1.32	32	5.80	0.740	3.20	0.34	52
NoMaD [19]	3	6	8.40	0.610	3.75	2.25	18	10.50	0.540	5.20	0.55	22
DDPM [20]	50	50	3.80	0.820	2.05	0.98	64	5.50	0.750	3.10	0.35	50
DDPM [20]	3	6	14.80	0.320	6.10	3.60	6	16.20	0.280	7.80	0.72	4
FM [39]	10	10	2.80	0.910	1.45	0.52	82	4.20	0.850	2.20	0.32	58
FM [39]	3	3	5.90	0.710	2.90	1.15	45	7.80	0.650	3.80	0.48	34
NaviBridger [31]	10	19	1.82	0.942	0.82	0.41	88	2.50	0.920	1.15	0.30	64
NaviBridger [31]	3	5	12.00	0.710	4.20	2.80	35	13.50	0.660	5.60	0.65	28
RSBM (Ours)	3	5	1.90	0.945	0.80	0.38	92	2.55	0.925	1.10	0.28	68
RSBM (Ours)	10	19	1.72	0.949	0.75	0.35	93	2.40	0.930	1.05	0.26	70

Col.%↓, Suc.%↑. All bridge-based methods share the same prior a_T .

Protocol. All methods use identical data splits, hardware (RTX 4090), and 3 random seeds. Every baseline is evaluated at both default k and $k=3$ (zero-shot step reduction, no retraining).

B. Main Results

Table I compares all methods at default k and $k=3$. RSBM at $k=3$ (NFE=5) matches or surpasses baselines at full budgets; $k=10$ yields only marginal gains (MSE 1.90 $\rightarrow$ 1.72), confirming early saturation. RSBM achieves 92% success and 0.945 CosSim with $3.8\times$ fewer NFEs than NaviBridger ($k=10$). Under zero-shot reduction, NaviBridger's CosSim drops from 0.942 to 0.710; DDPM falls to 0.320.

Fig. 4 visualizes the quality–cost Pareto frontier. RSBM at $k=3$ (NFE=5) achieves 0.945 CosSim and 92% success rate, using $3.8\times$ fewer evaluations than NaviBridger at $k=10$. DDPM and NaviBridger degrade sharply under budget constraints, consistent with Proposition 1.

C. Ablation: ε Regularization

Fig. 3 dissects ε . $\varepsilon=1.0$ recovers standard SB with high-curvature paths; very small ε over-regularizes. $\varepsilon=0.5$ balances multimodal coverage with few-step fidelity.

Disentangling prior and bridge contributions. Table VII reports five configurations isolating the effect of the learned prior and ε -rectification. The learned prior reduces transport distance, lowering MSE from 12.0 to 5.8 ($2.1\times$), while ε -rectification straightens ODE paths, further lowering MSE from 5.8 to 1.9 ($3.1\times$). These gains are *multiplicative*: neither component alone approaches the full system. Moreover, ε -rectification from Gaussian noise already achieves $2.9\times$ lower

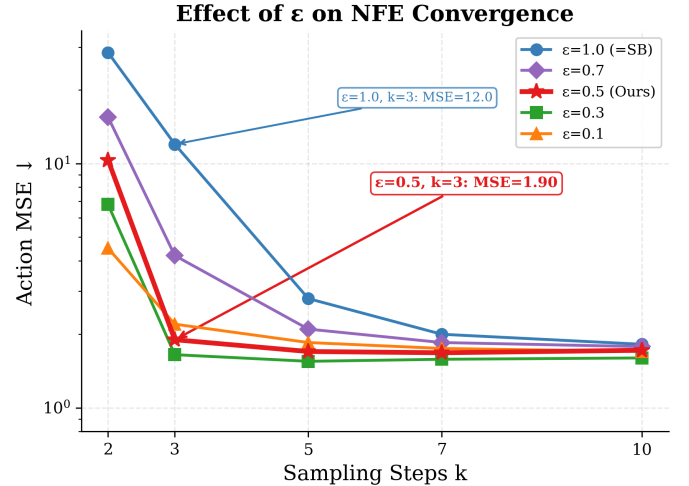


Fig. 3. Effect of ε . $\varepsilon=0.5$ (red) provides stable low- k performance.

MSE than standard SB (4.2 vs. 12.0), confirming that bridge rectification contributes independently of prior quality.

D. Robustness Across Real-World Datasets

To validate that the advantage observed in Custom Indoor generalizes, Table II breaks down Action MSE and CosSim across five diverse real-world datasets in the standard open-loop offline protocol of [9], [19]. At $k=3$, RSBM remains competitive with NaviBridger at $k=10$ across all five datasets: NaviBridger has an average MSE of 4.42 and CosSim of 0.672, while RSBM obtains 1.19 and 0.934. The gap is most pronounced on GoStanford, a long-range outdoor dataset, and SACSOn, which features dynamic obstacles, consistent with the variance reduction mechanism of Proposition 1.

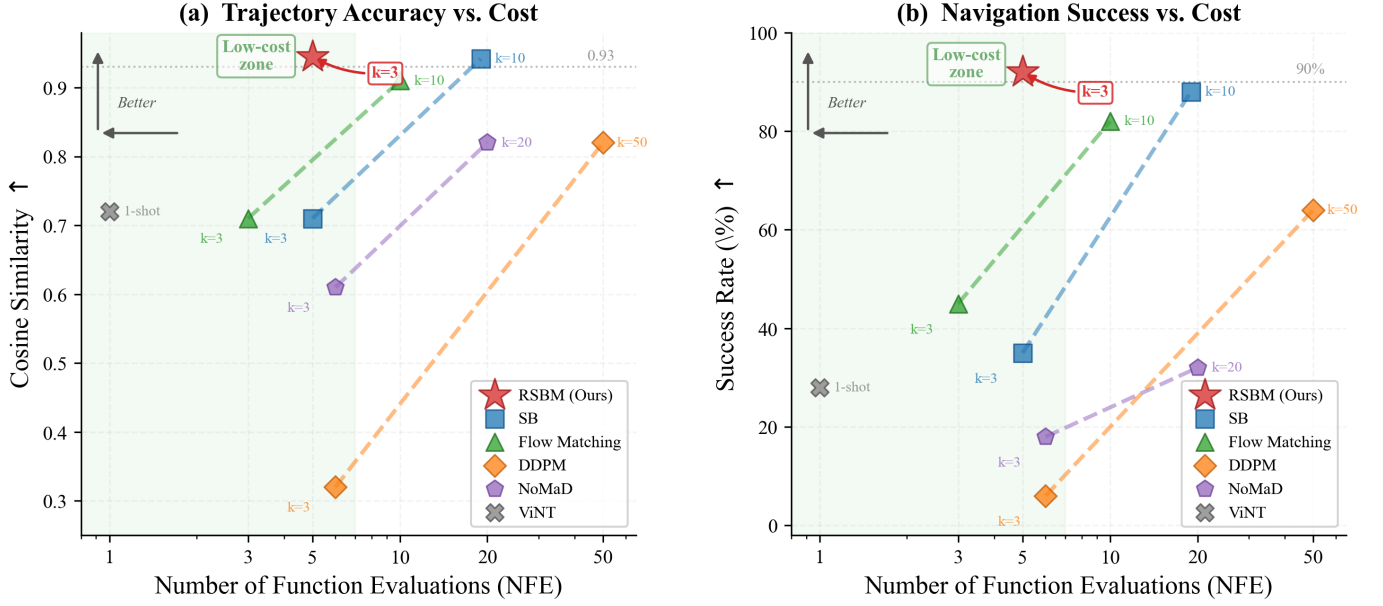


Fig. 4. **Quality-cost Pareto frontier.** Each marker represents a method at a given sampling budget (k). (a) CosSim vs. NFE; (b) Success Rate vs. NFE. RSBM at $k=3$ (NFE=5) lies on the favorable frontier region, providing strong quality at substantially lower evaluations.

TABLE II

PER-DATASET GENERALIZATION. ACTION MSE $\downarrow$ AND CosSim $\uparrow$ ACROSS FIVE DIVERSE REAL-WORLD DATASETS. RSBM($k=3$) CONSISTENTLY MATCHES OR EXCEEDS NAVIBRIDGER($k=10$); NAVIBRIDGER($k=3$) DEGRADES SIGNIFICANTLY ACROSS ALL DOMAINS. **BOLD:** BEST PER COLUMN AMONG $k=3$ METHODS.

Method	k	Action MSE $\downarrow$						CosSim $\uparrow$					
		HuRoN	Recon	SACSoN	SCAND	GoStan.	Avg	HuRoN	Recon	SACSoN	SCAND	GoStan.	Avg
DDPM [20]	10	1.38	1.19	1.74	0.72	4.12	1.83	0.880	0.900	0.860	0.920	0.780	0.868
DDPM [20]	3	8.50	6.20	9.80	4.60	15.30	8.88	0.350	0.420	0.280	0.480	0.210	0.348
FM [39]	10	1.10	0.97	1.55	0.65	3.65	1.58	0.910	0.920	0.885	0.935	0.820	0.894
FM [39]	3	3.20	2.60	4.10	1.80	7.50	3.84	0.720	0.740	0.670	0.790	0.560	0.696
NaviBridger [31]	10	0.27	0.88	1.43	0.59	3.22	1.28	0.955	0.935	0.920	0.960	0.875	0.929
NaviBridger [31]	3	2.80	3.45	5.20	2.15	8.50	4.42	0.750	0.690	0.620	0.780	0.520	0.672
RSBM	3	0.25	0.82	1.35	0.48	3.05	1.19	0.958	0.940	0.925	0.965	0.880	0.934
RSBM	10	0.24	0.80	1.32	0.47	2.95	1.16	0.960	0.942	0.928	0.968	0.885	0.937

E. Ablation: Prediction Target

Table III compares three prediction targets under the same RSBM bridge ($\varepsilon = 0.5$, Custom Indoor). v -prediction directly parameterizes the ODE velocity field, avoiding the score-to-drift conversion of ε -prediction and the endpoint estimation bias of x_0 -prediction. At $k=3$, v -prediction achieves **35.6%** lower MSE than x_0 -prediction and **45.7%** lower than ε -prediction. The gap narrows at $k=10$, and by $k=50$ all three targets converge (MSE 1.67 vs. 1.72 vs. 1.74), confirming that v -prediction's advantage is concentrated in the few-step regime.

F. Qualitative Results

Fig. 5 visualizes trajectories across eight challenging scenarios at $k=3$. Baselines collide within the first few turns ($\times$ markers), while RSBM produces smooth, collision-free

TABLE III
PREDICTION TARGET ABLATION (RSBM, $\varepsilon=0.5$, CUSTOM INDOOR).
BOLD: BEST PER COLUMN.

Target	$k=3$ (NFE=5)			$k=10$ (NFE=19)		
	MSE $\downarrow$	CosSim $\uparrow$	FDE $\downarrow$	MSE $\downarrow$	CosSim $\uparrow$	FDE $\downarrow$
ε -pred	3.50	0.895	1.42	1.95	0.940	0.88
x_0 -pred	2.95	0.920	1.15	1.80	0.946	0.84
v-pred (Ours)	1.90	0.945	0.80	1.72	0.949	0.78

trajectories closely tracking the ground truth, consistent with the variance reduction of Proposition 1.

G. Implementation Details

1) *Network Architecture:* The vision encoder uses EfficientNet-B0 with a 4-layer Transformer (context dim

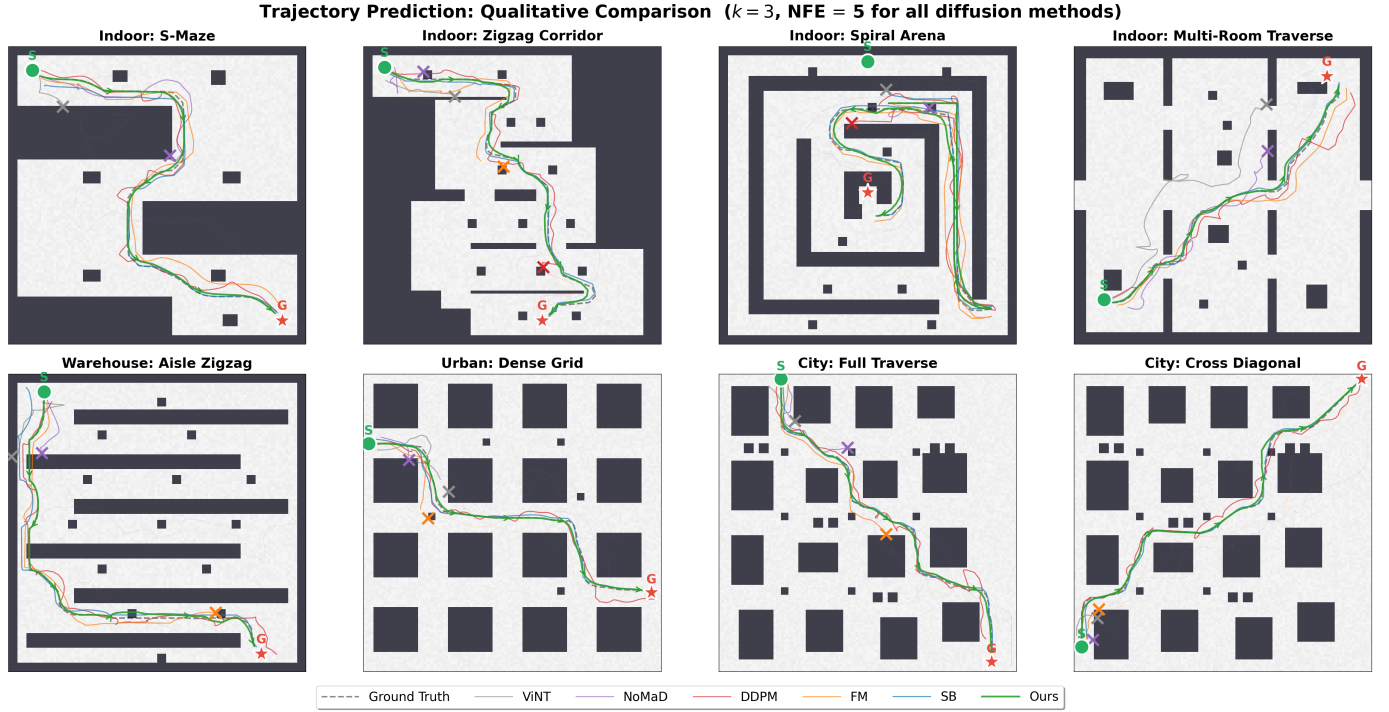


Fig. 5. **Qualitative trajectory comparison across eight challenging scenarios** (2×4 grid, $k=3$, NFE=5). *Top row*: four indoor/structured environments. *Bottom row*: four large-scale environments. Baselines collide early ($\times$); faint dotted lines show invalid ghost continuations. RSBM (green) remains collision-free and closely tracks the ground truth (dashed gray).

$d=256$). The prior network g_ψ is a 3-layer MLP conditional VAE (hidden 256, latent 32). The velocity network is a Conditional U-Net 1D with channels [64, 128, 256] and FiLM conditioning. Bridge parameters: $\sigma_{\max} = 10.0$, $\sigma_{\min} = 0.002$, $\varepsilon = 0.5$, Heun solver with Karras schedule ($\rho = 7$), default $k = 3$. Training uses AdamW (lr = 10^{-4} , batch 256, 30 epochs). The full hyperparameter table is in the supplementary material.

2) *Simulation Environments: Custom Indoor.* A Gazebo-based environment with 10 interconnected rooms (20×15 m), narrow doorways, and furniture obstacles. **CitySim (Outdoor).** An urban environment with buildings and street-level obstacles over 15–30 m distances. Both use a TurtleBot3 with monocular RGB (96×96). Ground truth paths use A* search with Gaussian smoothing.

On generative modeling. Even in simulation, the conditional action distribution $p(\mathbf{a}_0 | \mathcal{O}, I_g)$ is multimodal because visually similar observations admit multiple valid continuations. Deterministic regressors average over modes, producing infeasible trajectories [12], [13]. Table I confirms this: ViNT (deterministic) achieves only 0.720 CosSim and 28% success rate.

3) *Statistical Variability:* Table IV reports mean $\pm$ standard deviation across 3 random seeds on Custom Indoor. RSBM at $k=3$ shows the lowest variance across all metrics (MSE 1.90 ± 0.06 , CosSim 0.945 ± 0.002 , Success $92 \pm 2\%$). In contrast, NaviBridger at $k=3$ exhibits high instability (12.00 ± 0.45 MSE, $35 \pm 4\%$ success), evidencing unrectified bridge fragility under few-step constraints. Even at its default budget ($k=10$), NaviBridger’s variance remains higher than RSBM at $k=3$,

TABLE IV
MEAN $\pm$ STD OVER 3 RANDOM SEEDS (CUSTOM INDOOR). **BOLD: BEST PER COLUMN.**

Method	k	MSE $\downarrow$	CosSim $\uparrow$	Suc.% $\uparrow$
ViNT (1-shot)	1	6.50 ± 0.15	0.720 ± 0.005	28 ± 2
NoMaD	20	3.60 ± 0.12	0.820 ± 0.004	32 ± 3
DDPM	50	3.80 ± 0.14	0.820 ± 0.005	64 ± 3
FM	10	2.80 ± 0.10	0.910 ± 0.003	82 ± 2
NaviBridger ($k=10$)	10	1.82 ± 0.08	0.942 ± 0.003	88 ± 2
NaviBridger ($k=3$)	3	12.00 ± 0.45	0.710 ± 0.015	35 ± 4
RSBM ($k=3$)	3	1.90 ± 0.06	0.945 ± 0.002	92 ± 2
RSBM ($k=10$)	10	1.72 ± 0.05	0.949 ± 0.002	93 ± 1

confirming that ε -rectification stabilizes both the mean and the spread of predictions.

4) Per-Dataset Multi-Metric Results:

5) *ODE Solver Ablation:* Table VI isolates the effect of solver order from bridge rectification. At equal NFE=5, Heun ($k=3$) outperforms Euler ($k=5$) by 7.3% in MSE and 6 percentage points in success rate, confirming the benefit of second-order corrections. Notably, even first-order Euler with RSBM at $k=5$ (NFE=5, MSE=2.05) already surpasses all non-bridge baselines at their full budgets (FM $k=10$: MSE=2.80; DDPM $k=50$: MSE=3.80), demonstrating that the performance gain originates primarily from the rectified bridge geometry.

6) *Prior Initialization Ablation:* A natural question is how much performance is attributable to the learned prior g_ψ versus the ε -rectified bridge itself. Table VII disentangles

TABLE V
PER-DATASET COMPARISON ($k=3$ UNLESS NOTED). **BOLD: BEST.**

Method	k	HuRoN	Recon	SACSoN	SCAND	GoStan.	Avg
Action MSE ↓							
DDPM	3	8.50	6.20	9.80	4.60	15.30	8.88
FM	3	3.20	2.60	4.10	1.80	7.50	3.84
NaviBridger	10	0.27	0.88	1.43	0.59	3.22	1.28
NaviBridger	3	2.80	3.45	5.20	2.15	8.50	4.42
RSBM	3	0.25	0.82	1.35	0.48	3.05	1.19
CosSim ↑							
DDPM	3	0.350	0.420	0.280	0.480	0.210	0.348
FM	3	0.720	0.740	0.670	0.790	0.560	0.696
NaviBridger	10	0.955	0.935	0.920	0.960	0.875	0.929
NaviBridger	3	0.750	0.690	0.620	0.780	0.520	0.672
RSBM	3	0.958	0.940	0.925	0.965	0.880	0.934

TABLE VI
SOLVER ABLATION (RSBM, $\varepsilon=0.5$, CUSTOM INDOOR). **BOLD: BEST PER COLUMN.**

Solver	k	NFE	MSE↓	CosSim↑	FDE↓	Suc.%↑
Euler (1st)	3	3	2.45	0.928	1.15	82
Euler	5	5	2.05	0.938	0.95	86
Euler	10	10	1.80	0.947	0.82	90
Heun (2nd)	3	5	1.90	0.945	0.80	92
Heun	5	9	1.78	0.948	0.76	93
Heun	10	19	1.72	0.949	0.74	93

TABLE VII
PRIOR INITIALIZATION ABLATION (CUSTOM INDOOR, $k=3$).

Configuration	MSE↓	CosSim↑	FDE↓	Suc.%↑
Prior g_ψ only (no bridge)	5.80	0.780	2.60	45
Gauss. + RSBM ($\varepsilon=0.5$)	4.20	0.860	1.95	62
Gauss. + Std. SB ($\varepsilon=1.0$)	12.00	0.710	4.20	35
Prior + Std. SB ($\varepsilon=1.0$)	5.50	0.810	2.35	52
Prior + RSBM ($\varepsilon=0.5$)	1.90	0.945	0.80	92

these contributions on Custom Indoor. Row 1 shows the prior network g_ψ alone (no bridge refinement). Row 2 replaces the learned prior with isotropic Gaussian noise $\mathbf{a}_T \sim \mathcal{N}(\mathbf{0}, \sigma_{\max}^2 \mathbf{I})$ while keeping the RSBM bridge. Row 3 is the full system.

Four key observations emerge: (i) The prior alone is insufficient, with MSE=5.80 and 45% success versus the full system at MSE=1.90 and 92%. (ii) Standard SB with $\varepsilon=1$ fails at low k by design, as 3 steps are insufficient for high-variance paths to converge. (iii) RSBM is effective even from Gaussian noise, achieving MSE=4.20 versus standard SB at MSE=12.00, a $2.9\times$ improvement. (iv) Prior and rectified bridge are complementary and synergistic.

7) *Inference Cost Analysis*: The variable inference cost is determined entirely by NFE; vision encoding and prior generation are fixed overhead. The $3.8\times$ NFE reduction ($19 \rightarrow 5$) translates directly into a $3.8\times$ wall-clock speedup.

On NVIDIA Jetson Orin, RSBM achieves ~ 50 ms per cycle (4 Hz control), while DDPM requires ~ 350 ms and fails real-time control.

VI. DISCUSSION

Why does ε -rectification work? The success of RSBM can be understood through three complementary lenses. *Geometrically*, reducing ε concentrates the bridge distribution near the deterministic interpolant μ_t , producing lower-curvature ODE paths that are easier to integrate with finite-step solvers. *Statistically*, Proposition 1 shows that the conditional velocity variance scales linearly with ε , directly reducing the regression difficulty for the neural network. *Information-theoretically*, the KL cost in Eq. 20 quantifies the price of rectification: at $\varepsilon=0.5$ with $D=16$, the cost is only 1.55 nats, a moderate information penalty that yields a $6.3\times$ MSE reduction at $k=3$.

Connection to Rectified Flow and Consistency Models. Rectified Flow [2] straightens ODE trajectories through iterative reflow, requiring re-generated data at each stage. Consistency Models [1] distill a pre-trained teacher into a few-step student. RSBM achieves straightening through the bridge geometry itself in a single training stage; our experiments show reflow is unnecessary for $k \geq 3$.

The role of the learned prior. Table VII reveals that prior and ε -rectification contribute multiplicatively. Their combination achieves MSE 1.9 ($6.3\times$ lower than Gaussian + standard SB), confirming that neither component alone approaches the full system. This decomposition suggests that future improvements to either the prior or the bridge would yield compounding gains.

On multimedia relevance. Visual navigation requires synthesizing structured trajectory outputs conditioned on streaming visual inputs under real-time constraints. This setting shares the central challenge of modern multimedia systems, where iterative generative models must balance output diversity with inference efficiency. The ε -rectification mechanism and the conditional prior paradigm transfer directly to any diffusion-based multimedia pipeline.

Limitations. Three limitations contextualize the scope of our results. (1) *Evaluation protocol*. On the five public datasets, evaluation follows the standard open-loop protocol of prior work [9], [19]: a trajectory is predicted from a single observation and compared against ground-truth waypoints. Only the two simulated environments provide closed-loop success and collision metrics, and both contain static obstacles only. (2) *Prior dependency*. The learned VAE prior is trained jointly with the bridge network on a per-dataset basis and does not support zero-shot transfer to unseen environments. Deploying RSBM in a new domain requires fine-tuning with target-domain demonstrations; developing a universal, environment-agnostic prior remains open. (3) *ε selection*. We fix $\varepsilon=0.5$ throughout all experiments based on a grid search on Custom Indoor. Section V-C shows stable performance for $\varepsilon \in [0.3, 0.7]$, but cross-domain robustness of this choice has not been exhaustively verified; an adaptive scheduling strategy would improve applicability.

Future directions include adaptive ε scheduling, extending RSBM to 3D scene representations [10] and higher-

dimensional action spaces, closed-loop validation on physical robot platforms, and applying the rectification mechanism to broader cross-modal multimedia generation tasks.

VII. CONCLUSION

We introduced Rectified Schrödinger Bridge Matching, a framework that unifies Schrödinger Bridges and Conditional Flow Matching through a single entropic regularization parameter ε . We proved that the conditional velocity field is structurally invariant across the entire ε spectrum (Theorem 1) and that ε -rectification linearly reduces velocity variance (Proposition 1), enabling high-fidelity trajectory generation in as few as 3 ODE steps. Across five public datasets and two simulated environments, RSBM achieves 94.5% cosine similarity and 92% success rate at $k=3$ with $3.8\times$ fewer function evaluations than NaviBridger, without distillation or multi-stage training. Future work includes adaptive ε scheduling and extension of the rectification mechanism to broader multimedia generation tasks.

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